Aarya Gupta

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EDUCATION

Virginia Polytechnic Institute and State University — **Blacksburg, VA**
B.S. Computer Engineering | Dean's List — *Expected Dec 2028*

EXPERIENCE

Refresh (Y Combinator Backed AI Startup) — **San Francisco, CA**
Software Engineering Intern — *May 2026 – Aug 2026*

- Co-built a Dockerized clinical Electronic Health Record (EHR) system with responsive web UIs and 6+ workflows (e.g., patient intake) to benchmark AI computer-use agents on enterprise software.
- Engineered a multi-screen CUA evaluation framework, enabling AI research teams to measure cross-monitor navigation capabilities and agent tool-use reliability across complex desktop environments.
- Co-authored SoftValBench, a 20-task computer-use benchmark delivered to frontier labs and neo-labs to establish industry-standard evaluation metrics for desktop AI capabilities.
- Authored 30+ deterministic adversarial web tasks to systematically surface edge cases, security vulnerabilities, and logic failure modes in frontier computer-use models.
- Published 3 technical blog posts on refresh.dev covering trajectories.sh, CUA architecture, and evaluation benchmarks for the AI developer community.

Team Astra | NASA Micro-g NExT Challenge (Low Gravity Engineering Challenge) — **Blacksburg, VA**
Project Manager, Virginia Tech — *Sep 2025 – Present*

- Lead a 7-person engineering team designing a one-handed, passive-capture docking mechanism to overcome astronaut dexterity limits under stiff EVA gloves, meeting 100% of NASA design constraints.
- Drove CAD prototyping (Fusion 360) and 3D printing to iteratively test and engineer a high-reliability latching system optimized for low-gravity operations.
- Presented design specifications at a formal review to the Aerospace Engineering Department Head, incorporating faculty feedback into the final mechanical design.
- Co-authored technical proposals and safety documentation for NASA flight compliance; currently preparing a preprint paper for broader publication.

Harvard Medical School | Brigham and Women's Hospital — **Boston, MA**
Research Assistant — *Jul 2023 – Aug 2023; Jul 2024 – Aug 2024*

- Analyzed multi-modal MRI and PET scans of Multiple Sclerosis patients to quantify neuroinflammation markers for ongoing clinical research.
- Utilized DICOM imaging software to segment and trace brain lesions, generating longitudinal volume data used by clinicians to evaluate disease progression.
- Delivered a technical presentation on the foundational physics and mechanics of PET imaging systems to departmental research staff.

University of Illinois Urbana-Champaign | Dept. of Computer Science — **Urbana, IL**
Internet of Things (IoT) Developer — *Jun 2022 – Jul 2022*

- Built an IoT monitoring system using environmental sensors to gather real-time telemetry and automate plant growth forecasting, successfully deployed in a personal garden testbed.
- Implemented a CNN classifier to analyze crop images and distinguish fruits from vegetables, winning "Best Innovation" at a 24-hour hackathon.

SKILLS

Languages & Frameworks: C++, Python, Java, SQL, Bash, TypeScript, Node.js, Next.js
Infrastructure & AI Tools: Docker, Linux/Unix, Git, CI/CD, Anthropic API, AWS Bedrock, Gemini, GPT
Hardware & CAD: Arduino, Raspberry Pi, Fusion 360, AutoCAD, 3D Printing, Breadboarding
Lab & Medical Imaging: DICOM Medical Imaging, MRI/PET Data Analysis